Repair, Pair, Repack: Evidence-Level Context Compression for Long-Context Question Answering under Exact Token Budgets

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Abstract

Long-context question answering is often run under a hard prompt budget, imposed by cost, latency, or a model's context window: for Llama-3-8B, 59% of LongBench QA prompts exceed its 8K window. Popular prompt compressors meet such budgets by deleting tokens or phrases, and chunk retrievers by concatenating fixed-size chunks; both break evidence apart, either inside a sentence or at an arbitrary chunk boundary. We argue that compression for QA should operate on *evidence units* that stay verbatim and readable. We present BRIDGEPACK, a training-free context repacker that (i) selects query–support evidence pairs, (ii) repairs every selected span with its surrounding text so that answers cut by a boundary are restored, and (iii) admits evidence under the exact token count of the downstream model's own tokenizer before emitting it in document order. On all 1,150 English single- and multi-document QA instances of LongBench, with a 1,024-token budget (about 7–8$\times$ compression) and two LLMs, BRIDGEPACK outperforms LongLLMLingua by 6.6 F1, LLMLingua-2 by 11.3, Selective Context by 18.2, and head/middle truncation by more than 13, and it improves over BM25 and TF-IDF packing by 2.1 and 1.9 F1 (paired bootstrap, $p < 0.001$). It matches dense BGE retrieval without any encoder, and plugging dense recall into BRIDGEPACK yields the best compressed result on both models. Boundary repair alone contributes 2.4–2.7 F1 on each model, and the gains persist under nucleus sampling, across frozen hyperparameters, and on BABILong.

1 Introduction

Large language models (LLMs) are increasingly asked to answer questions over long documents and multi-document collections. Even when a model's context window is long enough, there are good reasons to send it less: inference cost and latency grow with prompt length, many deployed models still have short native windows, and long inputs dilute the few passages that actually matter (Liu et al., 2024a). In LongBench (Bai et al., 2023), for example, 59% of the English QA prompts are longer than the 8K-token window of Llama-3-8B-Instruct (Grattafiori et al., 2024), with a median of about 11K tokens. A practical system therefore needs to turn a long context into a short one under a *hard* budget, measured in the tokens of the model that will read it.

Two families of methods address this. *Prompt compressors* such as Selective Context (Li et al., 2023), LLMLingua (Jiang et al., 2023), LongLLMLingua (Jiang et al., 2024) and LLMLingua-2 (Pan et al., 2024) estimate the importance of tokens or phrases and delete the least informative ones. *Chunk retrievers* rank fixed-size chunks with BM25, TF-IDF or a dense encoder and concatenate the top ones (Robertson and Zaragoza, 2009; Karpukhin et al., 2020; Xiao et al., 2023b). Both treat compression as deciding *what to drop*. For extractive QA this has a cost that is easy to overlook: token deletion damages the sentences that carry the answer, and chunking cuts evidence at arbitrary boundaries, so an answer phrase can end up half inside and half outside the kept text. Multi-hop questions add a further difficulty, because the second-hop fact is often only weakly related to the question and is dropped by any ranker that scores spans one at a time.

We take a different view: under a strict budget, the unit of compression should be a *readable piece of evidence*. We present BRIDGEPACK, a training-free repacker built on three operations. **Pair**: candidate spans are selected as query–support pairs, where a span that answers the question directly is paired with a span that supports it, and the pair is scored jointly. **Repair**: every selected span is expanded with its neighbouring text, so that answer phrases cut by a span boundary are restored.

Repack: evidence is admitted with the downstream model's own tokenizer, so that the final context never exceeds the budget, and is emitted verbatim in document order. Figure 1 gives an overview.

We evaluate BRIDGEPACK under a protocol designed to make the comparison strict: all methods are evaluated with the same 1,024-token budget measured by the downstream tokenizer, the same prompt wrapper, model, decoding and scorer, on the full English single- and multi-document QA portion of LongBench (1,150 instances, six tasks), with two LLMs and paired bootstrap tests. Our findings are:

- **Evidence-level compression beats token-level compression by a wide margin.** BRIDGEPACK outperforms LongLLMLingua by 6.6 F1, LLMLingua-2 by 11.3 and Selective Context by 18.2 (two-model average, all $p < 0.001$), and the gaps persist on the three multi-hop tasks.
- **Pairing and repair make lexical evidence competitive with dense retrieval.** With the same lexical signals as BM25 and TF-IDF, BRIDGEPACK improves on them by 2.1 and 1.9 F1 ($p < 0.001$), matches dense BGE chunk retrieval without any encoder, and, when dense recall is plugged in (BRIDGEPACK-BGE), gives the best compressed result on both models.
- **Boundary repair is a key ingredient.** Removing it costs 2.7 F1 on Qwen and 2.4 on Llama-3, and removing the support term from pair scoring costs 1.4.
- **The gains are robust.** They hold under nucleus sampling, across one-at-a-time changes of every frozen constant, and on the synthetic BABILong benchmark, where BRIDGEPACK also uses 30% fewer prompt tokens than BM25.

2 Related Work

Prompt and context compression. Selective Context removes lexical units with low self-information under a small language model (Li et al., 2023). LLMLingua and LongLLMLingua use a small LM to estimate token importance, and LongLLMLingua adds question-aware coarse-to-fine compression for long contexts (Jiang et al., 2023, 2024). LLMLingua-2 distils token-level keep/drop labels into a bidirectional encoder (Pan et al., 2024). CPC trains a sentence encoder to score sentences against the question (Liskavets et al., 2025), and RECOMP and CompAct train extractive or abstractive compressors for retrieved documents (Xu et al., 2024; Yoon et al., 2024). BRIDGEPACK differs in three ways: it is training-free, it never edits the text inside a kept unit, and it selects evidence in query–support pairs rather than scoring units one at a time. We compare against the three published compressors that run off the shelf on arbitrary long contexts (Selective Context, LLMLingua-2, LongLLMLingua), all under the same exact budget.

Retrieval-style context construction. Retrieval-augmented generation ranks passages with sparse (Robertson and Zaragoza, 2009) or dense (Karpukhin et al., 2020; Xiao et al., 2023b) retrievers and concatenates the top results (Lewis et al., 2020; Izacard and Grave, 2021). Diversity-aware selection such as MMR (Carbonell and Goldstein, 1998) trades relevance against redundancy, and multi-hop retrieval chains retrieval steps over an external corpus (Xiong et al., 2021; Trivedi et al., 2023). BRIDGEPACK works inside a single given context, selects a supporting span *jointly* with the query-linked span it supports, and repairs span boundaries; its seed scorer is pluggable, and we show that a dense scorer and BRIDGEPACK's organisation are complementary.

Long-context evaluation and KV-cache compression. LongBench (Bai et al., 2023) and BABILong (Kuratov et al., 2024) evaluate question answering over long inputs, and models are known to under-use evidence placed in the middle of long contexts (Liu et al., 2024a). KV-cache methods reduce memory during generation by evicting, merging or quantising cached states (Xiao et al., 2023a; Zhang et al., 2023; Li et al., 2024; Liu et al., 2024b). BRIDGEPACK is complementary: it shortens the *input* before prefill, so the full long context is never encoded by the downstream model.

3 Method

3.1 Problem Setting

Given a question q and a long context C, we construct a compressed context $\hat{C}$ with $|\hat{C}|_\tau \le B$, where $|\cdot|_\tau$ counts tokens with the tokenizer τ of the downstream model and B is the budget. The task

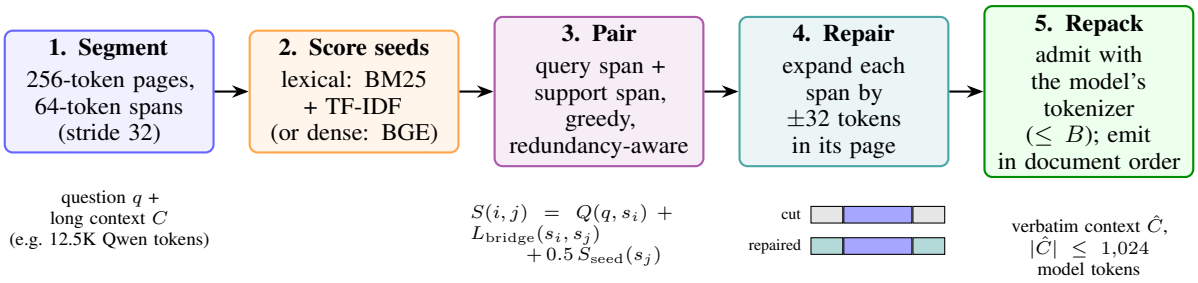


Figure 1: Overview of BRIDGEPACK. Spans are scored (lexically, or with a dense encoder in BRIDGEPACK-BGE), selected as query–support pairs, expanded with their neighbouring text so that boundary-cut evidence is restored, admitted under the exact budget of the downstream tokenizer, and emitted verbatim in document order. BRIDGEPACK runs before prefill; it shortens the input and does not modify the model or its KV cache.

instruction and the question are kept unchanged by a fixed prompt wrapper; only the context slot is replaced by $\hat{C}$. Because the budget is defined with the downstream tokenizer, a 1,024-token context for Qwen is a different text length from a 1,024-token context for Llama-3, and every method in our comparison is held to exactly the same definition.

3.2 Candidate Evidence Units

The context is split into contiguous, non-overlapping pages of 256 *regex tokens*, and each page into candidate spans of 64 regex tokens with stride 32; spans never cross page boundaries. Regex tokens are produced by the pattern

```
[A-Za-z0-9]+(?:['-][A-Za-z0-9]+)?|[^\w\s]
```

i.e., alphanumeric words with internal apostrophes or hyphens, or single punctuation characters. Regex tokens serve only to define candidate units independently of any model; all budget accounting uses the downstream tokenizer. All span- and chunk-based methods in our comparison operate on the same candidate spans.

3.3 Seed Scoring

Each span s receives a seed score. The default, encoder-free scorer fuses two lexical signals computed over the spans of the current instance,

$$S_{\text{seed}}(s) = \alpha\, z\big(S_{\text{BM25}}(q,s)\big) + (1-\alpha)\, z\big(S_{\text{TFIDF}}(q,s)\big), \quad (1)$$

where $z(\cdot)$ standardises scores within the instance, BM25 uses k_1=1.2, b=0.75, TF-IDF uses sublinear term frequency and smoothed IDF, and $\alpha = 0.5$ weights the two standardised signals equally. BRIDGEPACK-BGE replaces S_{seed} with the standardised cosine similarity between the question and the span under BGE (Xiao et al., 2023b); everything downstream is unchanged. The scorer never sees gold answers.

3.4 Query–Support Pair Selection

A span-by-span ranker keeps the spans that look most like the question. In multi-hop questions, however, the fact that completes the answer (e.g., a person's death date once the director is known) shares few words with the question and is ranked low. BRIDGEPACK therefore selects *pairs*. The seed pool holds up to 32 spans with positive query relevance $Q(q,s)$; the support pool holds up to 48 spans with positive seed score. A candidate pair (s_i, s_j) of a seed span and a support span is scored as

$$S(i,j) = Q(q,s_i) + L_{\text{bridge}}(s_i,s_j) + c_{\text{sup}}\, S_{\text{seed}}(s_j), \quad (2)$$

with $c_{\text{sup}} = 0.5$. The bridge term rewards lexical connection between the two spans while penalising near-duplicates,

$$L_{\text{bridge}}(i,j) = \frac{|T_i \cap T_j|}{\sqrt{|T_i|\,|T_j|}} + c_{\text{div}}\big(1 - J_{ij}\big) - c_{\text{jac}} J_{ij}, \quad (3)$$

where T_i is the content-term set of s_i, J_{ij} is their Jaccard overlap, $c_{\text{div}} = 0.10$ and $c_{\text{jac}} = 0.35$. Pairs are selected greedily, at most 24, by the redundancy-adjusted score $S(i,j) - \lambda\, J\big(T_i \cup T_j,\ T_{\text{sel}}\big)$ with $\lambda = 0.55$, where T_{sel} is the set of terms already covered; ties are broken by source position. In this way a weakly query-related span can enter the context because of the seed it supports, and pairs that add no new terms are discouraged.

3.5 Boundary Repair

Fixed-size spans cut sentences at arbitrary positions, and the cut often falls inside the answer

phrase (Table 4). BRIDGEPACK expands every selected span by up to 32 regex tokens on each side, clipped to its page, so that a repaired span covers up to 128 regex tokens.

3.6 Exact-Budget Repacking

Repaired spans are admitted in selection order while the packed context, measured with the downstream tokenizer τ, stays within B; admission stops at the first span that does not fit. The admitted spans are then emitted verbatim in their original document order, separated by blank lines, after the prefix "Question: q / Selected evidence:". If rejoining in document order changes the token count, the context is clipped to B tokens; this safeguard ensures $|\hat{C}|_\tau \leq B$ for every instance. Algorithm 1 in Appendix A summarises the procedure.

Hyperparameter protocol. All constants (α, pool sizes, c_{sup}, c_{div}, c_{jac}, λ, the repair radius) were fixed on a development split disjoint from the LongBench test instances before any test run, and are used unchanged for both models and all tasks. [TODO: one sentence naming the development split.] Section 5.5 shows that one-at-a-time changes to each of them never improve significantly on the frozen defaults.

4 Experimental Setup

Data. We use all English single- and multi-document QA tasks of LongBench (Bai et al., 2023): Qasper (Dasigi et al., 2021) (200), MultiFieldQA-en (150), HotpotQA (Yang et al., 2018) (200), 2WikiMultihopQA (Ho et al., 2020) (200), MuSiQue (Trivedi et al., 2022) (200) and NarrativeQA (Kočiský et al., 2018) (200), 1,150 instances in total. We report LongBench's task metric (token-level F1) and the macro average over the six tasks; "MH" is the average over the three multi-hop tasks.

Models and decoding. Qwen2.5-7B-Instruct (Qwen Team, 2024) and Meta-Llama-3-8B-Instruct (Grattafiori et al., 2024), with LongBench's task prompts inside each model's chat template, greedy decoding and LongBench's per-task output lengths. Section 5.5 repeats the main comparison under nucleus sampling.

Budget. The main budget is $B = 1{,}024$ downstream tokens for the context slot. Averaged per instance, this keeps 13.25% of the original prompt for Qwen and 13.80% for Llama-3 (7.5× and 7.2× compression; Appendix B).

Baselines. All baselines fill the same context slot under the same B and the same downstream tokenizer. *Truncation*: the first B tokens (head), or the first and last $B/2$ tokens (middle). *Published prompt compressors*, run with their official implementations: Selective Context (GPT-2, phrase level), LLMLingua-2 (`xlm-roberta-large-meetingbank`) and LongLLMLingua (question-aware, Llama-2-7B as the compressor LM). Each is asked to compress the context to the budget; its output is placed after the same "Question / Selected evidence" prefix as BRIDGEPACK, and if the result still exceeds B it is truncated in downstream tokens. *Span- and chunk-based packing* on the shared candidate spans: Span-only packing (top spans without repair), TF-IDF and BM25 chunk packing, and BGE chunk packing with `bge-small-en-v1.5` (query instruction, CLS pooling, top-64 spans, ±32-token neighbours). These are admitted under the same exact-budget rule. *Full context* is reported as a reference: Qwen reads the whole prompt (mean 12.5K tokens), while Llama-3 reads its 8K window with LongBench's middle truncation.

Statistics. Every comparison is paired on the same instances. We use a task-stratified paired bootstrap (Efron and Tibshirani, 1993) with 10,000 resamples and seed 20260701: instances are resampled within each task and the statistic is the macro average of per-task mean differences. Two-model results resample within the twelve model–task strata. We report 95% percentile intervals and two-sided bootstrap p-values.

5 Results

5.1 Evidence-level vs. token-level compression

Table 1 shows the main comparison and Table 2 the paired tests. Under an exact 1,024-token budget, all three published prompt compressors fall well below BRIDGEPACK: LongLLMLingua by 6.6 F1, LLMLingua-2 by 11.3 and Selective Context by 18.2 (two-model average, all $p < 0.001$). Selective Context is even below simple truncation. We see two reasons. First, deleting tokens or phrases damages exactly the material that extractive QA needs to copy: names, numbers and the function words that tie them together. BRIDGEPACK keeps every admitted unit verbatim. Second, at 7–8×

Family	Method	All six tasks			Multi-hop (MH)	
		Qwen	Llama-3	Avg.	Qwen	Llama-3
Reference	Full context (12.5K / 8K tokens)	43.30	36.73	40.02	44.91	36.13
Truncation	Head	25.47	23.92	24.70	27.05	22.90
	Middle	24.69	23.77	24.23	26.06	24.02
Prompt compression	Selective Context	20.15	19.13	19.64	22.02	20.67
	LLMLingua-2	26.82	26.31	26.57	30.57	28.86
	LongLLMLingua	32.64	29.87	31.26	36.95	31.42
Span / chunk packing	Span-only	22.31	20.52	21.42	25.15	20.76
	TF-IDF chunks	36.87	34.94	35.91	39.72	34.16
	BM25 chunks	36.94	34.44	35.69	39.85	33.94
	BGE chunks (dense)	39.65	36.75	38.20	44.04	37.36
Ours	BRIDGEPACK (encoder-free)	39.45	36.18	37.82	42.75	35.51
	BRIDGEPACK-BGE (dense seeds)	**39.89**	**37.19**	**38.54**	**44.09**	**37.64**

Table 1: Macro F1 on the 1,150 LongBench QA instances with a 1,024-token context budget measured by each model's tokenizer. MH averages HotpotQA, 2WikiMultihopQA and MuSiQue. Bold marks the best compressed method in each column. Per-task scores are in Appendix C.

BRIDGEPACK minus	All six tasks	Multi-hop
Selective Context	+18.18 [16.49, 19.88]	+17.79 [15.06, 20.40]
Middle truncation	+13.58 [11.92, 15.24]	+14.09 [11.54, 16.60]
Head truncation	+13.12 [11.43, 14.82]	+14.16 [11.56, 16.85]
LLMLingua-2	+11.25 [9.54, 12.97]	+9.41 [6.71, 12.14]
LongLLMLingua	+6.56 [4.99, 8.15]	+4.95 [2.47, 7.44]
Span-only	+16.40 [14.75, 18.06]	+16.18 [13.65, 18.69]
BM25 chunks	+2.12 [0.95, 3.31]	+2.24 [0.31, 4.20]
TF-IDF chunks	+1.91 [0.74, 3.11]	+2.19 [0.19, 4.14]

Table 2: Paired differences (F1) between BRIDGEPACK and each baseline, pooled over both models, with 95% bootstrap intervals. All intervals exclude zero ($p < 0.001$ on all six tasks; $p \leq 0.036$ on the multi-hop subset). Per-model intervals are in Appendix D.

compression the compressors must remove most of each sentence, whereas BRIDGEPACK can spend the whole budget on a few complete passages. The ordering is the same on the multi-hop subset, where BRIDGEPACK leads LongLLMLingua by 5.0 F1 and LLMLingua-2 by 9.4.

5.2 Pairing and repair vs. chunk retrieval

BM25 and TF-IDF chunk packing use the same lexical signals as BRIDGEPACK's seed scorer and the same candidate spans, so the difference isolates BRIDGEPACK's pair selection, boundary repair and repacking. BRIDGEPACK improves on BM25 by 2.12 F1 [0.95, 3.31] and on TF-IDF by 1.91 [0.74, 3.11], with gains of 2.2 F1 on the multi-hop tasks (Table 2). It improves on span-only packing, which selects spans one by one and does not repair them, by 16.4 F1.

Dense retrieval is a stronger baseline. Without any encoder, BRIDGEPACK is on par with BGE chunk packing: the paired difference is -0.38 F1 with an interval of $[-1.82, 1.05]$, i.e., the two are statistically indistinguishable. Because BRIDGEPACK's seed scorer is pluggable, the two signals can also be combined: BRIDGEPACK-BGE uses BGE similarities as seeds and BRIDGEPACK's pairing, repair and repacking. It is the best compressed method on both models and on both the full and the multi-hop task sets (Table 1), and improves on BM25 by 2.85 F1 [1.58, 4.14] and on TF-IDF by 2.64 [1.32, 3.98] (both $p < 0.001$; per model, all $p \leq 0.02$).

5.3 How close to the full context?

A 1,024-token BRIDGEPACK context recovers 91% of Qwen's full-context F1 (39.45 vs. 43.30) while sending about 8% of its prompt tokens (1,024 vs. a mean of 12.5K). For Llama-3, whose 8K window forces truncation of 59% of the full prompts, BRIDGEPACK with 1,024 tokens is statistically indistinguishable from the full 8K context (36.18 vs. 36.73; difference -0.54, 95% interval $[-2.85, 1.78]$, $p = 0.65$), and BRIDGEPACK-BGE scores above it (37.19). On the multi-hop tasks, BRIDGEPACK is within the bootstrap interval of the full context for both models. In other words, for these QA tasks most of the answer-relevant information in a long input can be packed into 1,024 verbatim tokens, provided the packing keeps evidence intact.

5.4 What drives the gains

Boundary repair. Disabling repair, with everything else unchanged, lowers F1 by 2.67 on Qwen

Removed component	Qwen	Llama-3	Two-model
Boundary repair	+2.67* [0.98, 4.43]	+2.36* [0.72, 4.02]	+2.52* [1.36, 3.71]
Support term (c_{sup}=0)	+1.86* [0.16, 3.60]	+0.86 [−0.86, 2.66]	+1.36* [0.14, 2.59]

Table 3: Ablations: F1 of the full BRIDGEPACK minus the ablated variant (paired bootstrap, 95% intervals; *: interval excludes zero). Each row changes one component and keeps everything else, including the exact budget, fixed.

Q: Where did they film *The Beach* in the country where Siddhi Savetsila was born? **Gold:** Ko Phi Phi Le

BRIDGEPACK: *"... The film stars Leonardo DiCaprio, ... It was filmed on the Thai island of Ko Phi Phi Le."* → **Ko Phi Phi Le** ✓

w/o repair: the span ends at *"... landscaping of the natural beach setting of Ko Phi Phi"*; the sentence naming the filming location is not in the context → *Thailand* ×

BM25 chunks → *They filmed The Beach in Thailand.* ×
BGE chunks → *Khao Yai National Park in central Thailand* ×

Table 4: A MuSiQue example (Qwen, 1,024 tokens). Repair restores the sentence that names the filming location; without repair, the selected span ends mid-phrase and the model falls back to the country.

and 2.36 on Llama-3 (Table 3); both intervals exclude zero. The same holds for a variant without lexical seeding (Qwen +2.79 [1.09, 4.43], Llama-3 +2.13 [0.42, 3.88]). Repair costs budget, since each repaired span is up to twice as long, so this gain is a net effect: under a fixed budget, fewer complete pieces of evidence are better than more fragments. Table 4 shows a typical case.

Support term. Setting $c_{\text{sup}} = 0$ reduces a pair's score to the relevance of its seed plus the bridge term, so supporting spans are no longer chosen for their own evidence value. This lowers F1 by 1.36 [0.14, 2.59] across the two models (1.86 on Qwen).

5.5 Robustness

Decoding. BRIDGEPACK changes only the input text, so its benefit should not depend on greedy decoding. With nucleus sampling (Holtzman et al., 2020) ($T = 0.7$, top-$p = 0.9$) and three seeds on Qwen, BRIDGEPACK improves on BM25 by 2.78 F1 [1.19, 4.43] on the seed-averaged scores, and by 2.74, 2.81 and 2.77 for the individual seeds (each $p \leq 0.002$).

Frozen constants. Table 8 (Appendix E) changes one constant at a time around the frozen defaults: the fusion weight $\alpha \in \{0.25, 0.75\}$, the support weight $c_{\text{sup}} \in \{0.25, 1.0\}$, the redundancy weight $\lambda \in \{0.30, 0.70\}$, the bridge constants $c_{\text{div}} = 0$ and $c_{\text{jac}} = 0$, and the repair radius $\in \{16, 64\}$. Every change moves macro F1 by at most 0.43 on Qwen and 0.72 on Llama-3, apart from the repair radius, and no setting is significantly better than the default. The default radius of 32 is the best radius on Qwen and within 0.06 F1 of the best on Llama-3; doubling it to 64 costs 1.2–1.5 F1 because fewer repaired spans fit in the budget.

Synthetic long-context reasoning. On BABILong (Kuratov et al., 2024) (tasks qa1–qa3 at 16K and 32K tokens, 200 instances each, 1,024-token budget, Qwen), BRIDGEPACK reaches 32.17 accuracy vs. 29.25 for BM25 chunk packing, a paired gain of 2.92 [1.17, 4.67] ($p < 0.001$), while its prompts are 30% shorter on average (763 vs. 1,087 tokens including the task wrapper).

6 Discussion

Why compress at the level of evidence? Token-level compressors were designed for settings where a moderate compression ratio leaves most of each sentence intact. Under a strict budget of 7–8× they must delete most of the text, and the result is no longer readable evidence. Chunk retrievers keep text verbatim but cut it at positions that ignore its content. BRIDGEPACK combines the strengths of both: like a retriever, it keeps evidence verbatim, and like a compressor, it decides what each unit should contain, here by repairing its boundaries. The ablations suggest that this second decision matters as much as the selection itself.

Lexical or dense seeds? The encoder-free BRIDGEPACK needs nothing beyond term statistics of the current context and is on par with a dense retriever. When a dense encoder is available, BRIDGEPACK-BGE is the strongest compressed method in our study. The two instantiations therefore offer a simple choice between zero additional models and the best accuracy.

Deployment. BRIDGEPACK runs on the CPU before the model is called, needs no training, and requires only the downstream tokenizer to enforce the budget. Because the output is verbatim source text in document order, the compressed context remains inspectable by users, which matters in settings where answers must be traced back to their sources.

7 Conclusion

We studied long-context question answering under a hard budget measured in the downstream model's own tokens. Compressing at the level of readable evidence units—selected in query–support pairs, repaired at their boundaries and admitted under the exact budget—outperforms token-level prompt compressors by 6.6–18.2 F1 and lexical chunk retrieval by about 2 F1, matches dense retrieval without an encoder, and with dense seeds gives the best compressed results on two LLMs. At 1,024 tokens it recovers 91% of Qwen's full-context accuracy and matches Llama-3's full 8K context. We hope this encourages evaluating context compression not only by *what* it keeps, but by whether what it keeps is still usable evidence.

Limitations

BRIDGEPACK shortens the input before prefill; it does not reduce the memory needed to encode a long input that has already been sent to the model, and it is not a KV-cache method. Our experiments cover English extractive QA on LongBench and BABILong with two 7–8B instruction models; other languages, generation tasks such as summarisation, and larger models may behave differently. We study a budget of 1,024 tokens, a 7–8$\times$ compression that is large enough to hold several repaired pairs; when the budget is so small that a single repaired span fills it, pairing has no room to operate. The dense instantiation requires an additional encoder at inference time. Finally, the lexical scorer relies on term overlap; questions that share no vocabulary with their evidence are better served by the dense seeds.

Ethics Statement

Context compression can remove information that a user would consider important, which can lead to incomplete or incorrect answers; in high-stakes settings, compressed contexts should be shown to the user or checked against the source. BRIDGEPACK keeps all admitted text verbatim, which makes such checks straightforward. All datasets are public research benchmarks used under their licences.

References

Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2023. Longbench: A bilingual, multitask benchmark for long context understanding. *arXiv preprint arXiv:2308.14508*.

Jaime Carbonell and Jade Goldstein. 1998. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336.

Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. 2021. A dataset of information-seeking questions and answers anchored in research papers. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4599–4610.

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman and Hall/CRC.

Aaron Grattafiori et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration. In *Proceedings of the International Conference on Learning Representations*.

Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 874–880.

Huiqiang Jiang, Qianhui Wu, Chin-Yew Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2023. Llmlingua: Compressing prompts for accelerated inference of large language models. *arXiv preprint arXiv:2310.05736*.

Huiqiang Jiang, Qianhui Wu, Chin-Yew Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2024. Longllmlingua: Accelerating and enhancing llms in long context scenarios via prompt compression. *arXiv preprint arXiv:2310.06839*.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781.

Tomáš Kočiský, Jonathan Schwarz, Phil Blunsom, Chris Dyer, Karl Moritz Hermann, Gábor Melis, and Edward Grefenstette. 2018. The NarrativeQA reading comprehension challenge. *Transactions of the Association for Computational Linguistics*, 6:317–328.

Yuri Kuratov, Aydar Bulatov, Petr Anokhin, Ivan Rodkin, Dmitry Sorokin, Artyom Sorokin, and Mikhail Burtsev. 2024. BABILong: Testing the limits of LLMs with long context reasoning-in-a-haystack. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Yucheng Li, Bo Dong, Chenghua Lin, and Frank Guerin. 2023. Selective context: Efficient context selection for large language models. *arXiv preprint arXiv:2304.12102*.

Yucheng Li, Ying Huang, Bowen Yang, Bharat Venkitesh, Andrea Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. Snapkv: Llm knows what you are looking for before generation. *arXiv preprint arXiv:2404.14469*.

Barys Liskavets, Maxim Ushakov, Shuvendu Roy, Mark Klibanov, Ali Etemad, and Shane K. Luke. 2025. Prompt compression with context-aware sentence encoding for fast and improved LLM inference. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 24595–24604.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024a. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. 2024b. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. *arXiv preprint arXiv:2402.02750*.

Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor Rühle, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao, Lili Qiu, and Dongmei Zhang. 2024. LLMLingua-2: Data distillation for efficient and faithful task-agnostic prompt compression. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 963–981.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 10014–10037.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2023a. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*.

Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. 2023b. C-pack: Packaged resources to advance general Chinese embedding. *arXiv preprint arXiv:2309.07597*.

Wenhan Xiong, Xiang Lorraine Li, Srinivasan Iyer, Jingfei Du, Patrick Lewis, William Yang Wang, Yashar Mehdad, Wen-tau Yih, Sebastian Riedel, Douwe Kiela, and Mohit Bansal. 2021. Answering complex open-domain questions with multi-hop dense retrieval. In *Proceedings of the International Conference on Learning Representations*.

Fangyuan Xu, Weijia Shi, and Eunsol Choi. 2024. Recomp: Improving retrieval-augmented lms with compression and selective augmentation. In *Proceedings of the International Conference on Learning Representations*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

Chanwoong Yoon, Taewhoo Lee, Hyeon Hwang, Minbyul Jeong, and Jaewoo Kang. 2024. CompAct: Compressing retrieved documents actively for question answering. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21424–21439.

Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Re, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023. H2o: Heavy-hitter oracle for efficient generative inference of large language models. *arXiv preprint arXiv:2306.14048*.

A Implementation Details

BRIDGEPACK-BGE uses the same pair selection and repair as BRIDGEPACK; its spans are selected with regex-token budget estimates, re-admitted under the downstream tokenizer, and emitted in selection order. The code, packed contexts and per-instance predictions for every number in this paper will be released.

Algorithm 1 BRIDGEPACK

Require: question q, context C, budget B, tokenizer τ
split C into 256-token pages and 64-token spans (stride 32)
$S_{\text{seed}} \leftarrow$ fused BM25/TF-IDF z-scores (or BGE)
seeds $\leftarrow$ top 32 spans with $Q(q, s) > 0$; supports $\leftarrow$ top 48 with $S_{\text{seed}} > 0$
$\mathcal{P} \leftarrow \emptyset$, $T_{\text{sel}} \leftarrow \emptyset$
while $|\mathcal{P}| < 24$ and an unselected pair remains **do**
    pick (s_i, s_j) maximising $S(i, j) - \lambda J(T_i \cup T_j, T_{\text{sel}})$
    add (s_i, s_j) to $\mathcal{P}$; $T_{\text{sel}} \leftarrow T_{\text{sel}} \cup T_i \cup T_j$
end while
repair each span in $\mathcal{P}$ by ± 32 tokens within its page
admit repaired spans in selection order while $|\hat{C}|_\tau \leq B$
emit admitted spans in document order; clip to B if needed
return $\hat{C}$

Item	Value
Pages / spans	256 regex tokens / 64, stride 32
Seed fusion	$0.5\, z(\text{BM25}) + 0.5\, z(\text{TFIDF})$
BM25 / TF-IDF	k_1=1.2, b=0.75 / sublinear TF, smoothed IDF
Seed / support pools	32 / 48 spans
Max. pairs	24
c_{sup}, c_{div}, c_{jac}, λ	0.5, 0.10, 0.35, 0.55
Repair radius	32 regex tokens per side, page-clamped
Admission	downstream tokenizer, stop at first non-fit
Emission order	document order
Dense seeds (BRIDGEPACK-BGE)	`bge-small-en-v1.5`, CLS pooling, ℓ_2-normalised, query instruction prefix

Table 5: Implementation constants (all frozen before testing).

B Compression Ratios

The effective ratio of an instance is $\min(B, T)/T$, where T is the length of the full prompt under the downstream tokenizer. Averaged over the 1,150 instances, a 1,024-token budget keeps 13.25% of the prompt for Qwen (7.55× compression) and 13.80% for Llama-3 (7.25×). Under the Llama-3 tokenizer, 682 of the 1,150 full prompts (59%) exceed 8,192 tokens.

C Per-Task Results

Table 6 lists per-task F1 for all methods in Table 1.

D Per-Model Paired Tests

Table 7 reports the paired comparisons of Table 2 separately for each model, together with those of BRIDGEPACK-BGE.

E Sensitivity to Frozen Constants

Model	Method	Qasper	MF-en	HotpotQA	2Wiki	MuSiQue	NarrQA
Qwen	Full context	43.61	52.60	57.68	47.19	29.85	28.89
	Head trunc.	24.77	38.39	32.83	31.59	16.74	8.53
	Middle trunc.	22.56	35.46	34.18	28.93	15.07	11.95
	Selective Context	24.07	22.77	27.66	28.26	10.14	7.97
	LLMLingua-2	28.01	30.31	37.34	36.54	17.84	10.89
	LongLLMLingua	31.20	39.84	48.12	39.88	22.85	13.95
	Span-only	16.86	32.57	28.53	29.50	17.41	8.97
	TF-IDF chunks	36.99	49.08	52.38	40.21	26.59	15.95
	BM25 chunks	38.10	49.33	52.41	41.00	26.13	14.69
	BGE chunks	40.10	47.80	52.52	48.10	31.50	17.88
	BRIDGEPACK	41.76	49.89	53.91	46.33	28.02	16.76
	BRIDGEPACK-BGE	41.23	48.64	55.00	49.90	27.36	17.19
Llama-3	Full context (8K)	42.98	48.06	46.99	39.10	22.29	20.95
	Head trunc.	26.95	36.85	33.07	22.86	12.77	11.01
	Middle trunc.	24.14	33.55	32.49	26.10	13.48	12.85
	Selective Context	23.95	19.58	28.36	23.17	10.48	9.23
	LLMLingua-2	31.87	26.00	37.02	31.31	18.25	13.40
	LongLLMLingua	31.04	38.96	42.27	31.39	20.59	14.97
	Span-only	19.34	30.49	27.66	23.07	11.54	11.04
	TF-IDF chunks	39.40	49.63	47.04	34.36	21.09	18.09
	BM25 chunks	38.10	49.47	49.48	32.76	19.58	17.24
	BGE chunks	40.84	47.52	46.08	39.08	26.92	20.03
	BRIDGEPACK	39.94	50.35	47.06	39.73	19.74	20.27
	BRIDGEPACK-BGE	42.22	50.29	49.89	38.33	24.68	17.73

Table 6: Per-task F1 at a 1,024-token budget.

Comparison	Qwen	Llama-3	Two-model
BRIDGEPACK − Selective Context	+19.30 [16.90, 21.71]	+17.06 [14.57, 19.49]	+18.18 [16.49, 19.88]
BRIDGEPACK − Middle truncation	+14.75 [12.37, 17.13]	+12.41 [10.07, 14.78]	+13.58 [11.92, 15.24]
BRIDGEPACK − Head truncation	+13.97 [11.58, 16.37]	+12.27 [9.96, 14.67]	+13.12 [11.43, 14.82]
BRIDGEPACK − LLMLingua-2	+12.62 [10.24, 15.07]	+9.88 [7.47, 12.32]	+11.25 [9.54, 12.97]
BRIDGEPACK − LongLLMLingua	+6.81 [4.59, 9.05]	+6.31 [4.01, 8.66]	+6.56 [4.99, 8.15]
BRIDGEPACK − Span-only	+17.14 [14.79, 19.52]	+15.66 [13.34, 18.04]	+16.40 [14.75, 18.06]
BRIDGEPACK − BM25 chunks	+2.50 [0.78, 4.26]	+1.75 [0.13, 3.42]	+2.12 [0.95, 3.31]
BRIDGEPACK − TF-IDF chunks	+2.58 [0.87, 4.33]	+1.25 [−0.32, 2.89]	+1.91 [0.74, 3.11]
BRIDGEPACK-BGE − BM25 chunks	+2.94 [1.04, 4.80]	+2.75 [0.87, 4.60]	+2.85 [1.58, 4.14]
BRIDGEPACK-BGE − TF-IDF chunks	+3.02 [1.14, 4.87]	+2.26 [0.37, 4.21]	+2.64 [1.32, 3.98]
BRIDGEPACK-BGE − Span-only	+17.58 [15.14, 19.99]	+16.67 [14.29, 19.04]	+17.12 [15.40, 18.83]

Table 7: Per-model paired bootstrap differences (F1, 95% intervals) on all six tasks.

Setting (default)	Qwen	Llama-3
Default	39.45	36.18
$\alpha = 0.25$ (0.5)	39.22	36.20
$\alpha = 0.75$	39.19	35.95
$c_{\text{sup}} = 0.25$ (0.5)	39.01	36.90
$c_{\text{sup}} = 1.0$	39.37	35.75
$\lambda = 0.30$ (0.55)	39.02	36.21
$\lambda = 0.70$	39.35	36.04
$c_{\text{div}} = 0$ (0.10)	39.43	–
$c_{\text{jac}} = 0$ (0.35)	39.21	–
Repair radius 16 (32)	38.84	36.24
Repair radius 64	37.99	35.02

Table 8: One-at-a-time changes of the frozen constants (macro F1, 1,024 tokens). No setting is significantly better than the default (paired bootstrap).